Joshua Su

770-309-4820 | joshua.su@gatech.edu | joshuasu7.github.io | US Citizen

Education

Georgia Institute of Technology August 2024 – Expected May 2028
Bachelors of Science in Electrical Engineering, GPA: 3.85 *Atlanta, GA*

Skills

Engineering | RADAR, Antenna Engineering, RF Design, PCB Design, Signal Processing, Embedded Systems, FPGAs, Oscilloscopes, Logic Analyzers

EDA & Simulation | ANSYS Electronics Desktop, Altium Designer, Altair FEKO, Cadence Virtuoso, Altera Quartus Prime

Programming | Python, C/C++, MATLAB, VHDL

Personal Projects

X-Band Phased Array RADAR Board July 2026 - Present
Antenna & PCB Design | ANSYS Electronics | Altium Designer

- Designed a 10.5GHz 4x4 Microstrip patch antenna phased array in ANSYS Electronics, achieving a peak realized gain of 17.3dBi at boresight, 3dB beamwidth of 13°, and an S_{11} return loss of -24dB at 10.5GHz with a 700MHz (-10dB) bandwidth
- Achieved $\pm 60^\circ$ electronic beamsteering with $\leq$ 3dB scan loss, validated by sweeping element phase excitations and evaluating realized gain across θ and ϕ scan planes
- Built 8-layer RF module PCB in Altium Designer integrating the array, 4 ADAR1000 beamformers for 16-element phase control, an SDR, and an ESP32 handling SPI control of the LO and beamformers; RF chain includes a TX/RX-shared transverter (TCXO, LO, mixer), BPFs, PAs, LNA, attenuators, and T/R switches, with the stackup isolating RF, digital, and power domains
- Designed impedance-matched, length-matched GCPW and symmetrical stripline traces for high-speed RF signal routing, minimizing crosstalk and coupling while eliminating phase delay discrepancies between beamformer channels

Experience

Antenna/RF Design Engineer — GT Experimental Rocketry - Ground Systems August 2026 – Present
Antenna Design | Altair FEKO

- Engineering a two-way RF ground station system for rocket-to-mission-control telemetry, designed to eliminate prior-launch communication dropouts with a modeled +58dB link margin
- Validated antenna link performance ahead of fabrication, confirming far-field radiation patterns met mission link-budget requirements through full rocket 3D mesh and test-site topology simulations in Altair FEKO
- Determined external RF hardware architecture (receive antenna, 19dB-gain LNA, bias tee injector, SX1262 LoRa transceiver (433MHz/62.5kbps), Raspberry Pi, WiFi point-to-point module)

Hardware Engineer — GT Yellow Jacket Space Program - Avionics August 2026 - Present
PCB Design | Altium Designer

- Designed a 6-channel avionics sensor interface board in Altium Designer, implementing a 24V input protection stage (fuse, TVS diode, bulk capacitance) and a buck/LDO power tree delivering regulated 5V and 3.3V rails to downstream digital and analog circuitry
- Developed a hierarchical schematic and SPI-based signal chain connecting an STM32 microcontroller to an external ADC, enabling digitization of six 4-20mA current-loop sensor channels for pressure, temperature, and force measurement applications
- Laid out PCB with room-based floorplanning, sizing power-rail trace widths for current capacity, isolating short analog sensor traces from noisy high-current paths, and stitching ground-pour copper for signal integrity

Software Developer, Electrical Engineer — GT Medical Robotics Club August 2025 – May 2026
Signal Processing | Python

- Built electrooculography (EOG)-based eye tracking software for smart glasses that act as a human-computer interface for individuals with locked-in syndrome
- Developed real-time eye tracking and blink detection algorithm from onboard EOG electrode signals, achieving 90% blink detection accuracy across $\sim$ 20 participants and sub-100ms cursor response latency in the processing chain
- Applied custom calibration technique and signal differencing pipeline to extract clean signals despite significant person-to-person variability in baseline signal intensity and noise

Relevant Coursework

RADAR and Electromagnetic Sensing, Electromagnetics and Microwave Applications, Electromagnetics, Signals and Systems, Signal Processing, Microelectronic Circuits, Programming HW/SW Systems, Digital Design Lab